Lesson 6.8 Guided Notes

Session 1 · Using Properties to Rewrite Expressions

I can use the distributive property and like terms to rewrite an expression in simplest form. · 6.EE.A.3

Word / Spanish	Meaning — complete the blank	Example
Distributive Property / propiedad distributiva	Multiply the outside number by _____ term inside the parentheses.	a(b + c) = ab + ac
Commutative Property / propiedad conmutativa	Changing the _____ when adding or multiplying does not change the answer.	a + b = b + a
Associative Property / propiedad asociativa	Changing the _____ when adding or multiplying does not change the answer.	(a + b) + c = a + (b + c)
like terms / términos semejantes	Terms with the same _____ part.	4m and 3m

Key idea

Expressions are _____ when they give the same value for every value of the variable.

To simplify: distribute, put like terms together, then _____ them.

Worked example

Simplify 4(m + 2) + 3m.

Distribute: 4(m + 2) = 4m + _____.

Reorder: 4m + 3m + 8 (Commutative Property).

Combine like terms: 4m + 3m = _____.

Answer: 4(m + 2) + 3m = _____

You try

1. Write each expression in simplest form.
a. 2(x + 5) = _____
b. 3(y + 1) + 2y = _____

_____

2. Simplify 6(c + 2) + c + 4. Check your answer with c = 1.

_____

Sum it up: Two expressions are equivalent when ___, so I can check by ___.

_____

Session 2 · Expanding and Factoring Expressions

I can expand an expression and factor one using the greatest common factor. · 6.EE.A.3

Word / Spanish	Meaning — complete the blank	Example
expand / desarrollar	Multiply to _____ the parentheses.	6(3x + 4) = 18x + 24
factor an expression / factorizar una expresión	Write it as a _____: a shared factor times a sum.	18x + 24 = 6(3x + 4)
term / término	One _____ of an expression, separated by + or − signs.	18x and 24 are terms
greatest common factor / máximo común divisor	The _____ number that divides both numbers evenly.	GCF of 18 and 24 is 6

Key idea

Factoring _____ expanding, so you can check a factored form by expanding it.

Factor out the _____ so the terms inside share no other common factor.

Worked example

Factor 18x + 27 using the GCF.

The GCF of 18 and 27 is _____.

Divide each term: 18x ÷ 9 = _____ and 27 ÷ 9 = 3.

Check: 9 × 2x + 9 × 3 = 18x + 27.

Answer: 18x + 27 = _____

You try

1. Expand or factor.
a. Expand 4(2x + 3) _____
b. Factor 15y + 20 _____

_____

2. Factor 16n + 24 using the GCF. Then expand to check.

_____

Sum it up: To factor an expression, I ___, and I check by ___.

_____

Session 1 · Model, reason, and explain

Using Properties to Rewrite Expressions · 6.EE.A.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use a partitioned rectangle to expand 3(x+4), then simplify 3(x+4)+2x. Explain which property justifies each change.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: distributes to only the variable term.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Expanding and Factoring Expressions · 6.EE.A.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Factor 12x+18 using the greatest common factor. Expand to verify. Is 2(6x+9) also equivalent, and why is it not factored by the GCF?

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: accepts any common factor as the greatest or fails to divide every term.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?